

Element A: Presentation and Justification of the Problem

Our school produces a lot of paper waste. A large majority of this waste comes from sheets of paper that have only been used or printed on one side. The other side is blank and completely usable, but the paper is getting thrown out anyway. This leads to more cost for the school since they have to pay more money for paper. It also led to more work for our housekeepers, who have to haul all of the extra paper waste out of the building and into dumpsters. The other paper waste comes from the composition notebooks that students use for classes. The cost of these notebooks comes directly out of each teacher's budget. However, in most cases, less than the first third of the notebook is used, and then it is not used again and just gets thrown out, even though the other two-thirds of the notebook have never been used.

Successfully solving this problem could significantly impact the stakeholders in a positive way. For example, it could save our stakeholders a lot of money. An unintended consequence of solving our problem is that our stakeholders would have to find someone or a team of people to continue building new notebooks from old ones.

Our primary stakeholders are Mr. B and Mr. F. Mr. B is the technology coordinator at our school. While Mr. F is the director of facilities at our school. Our school's students, alumni, teachers, and faculty were also stakeholders. If the school spent less money on paper and notebooks, it would be able to use that money for other things that would benefit the school. This problem likely occurs in many schools and may occur in offices or other places that use a lot of paper.

Our school uses around 280,000 pages of paper per school year. Schools in the United States, on average, use 320,000 sheets of paper per school year. Across the

approximately 100,000 schools in the United States, that means schools in the United States alone use a total of around 32 billion sheets of paper per school year. (McHugh, 2026) Additionally, around 30% of household waste by weight was paper; the average American uses 700 lb of paper each year. This is the highest paper use per capita in the world. ("Identifying Greener Paper | US EPA"). This shows that this was not just an issue related to school; it was a national, and most likely global, problem.

It was important to solve this problem now because the school continues to waste money on paper every day. This was a problem that needed to be solved since it saved the school money and reduced paper waste.

References

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